PRACTICE SHEET-1

1. If the height of a given cone be doubled and radius of the base remains the same, the ratio of the volume of the given cone to that of the second cone will be
(1) 2 : 1 (2) 1 : 8
(3) 1 : 2 (4) 8 : 1

2. If the radius of the base of a cone be doubled and height is left unchanged, then ratio of the volume of new cone to that of the original cone will be:
(1) 1 : 4 (2) 2 : 1
(3) 1 : 2 (4) 4 : 1

3. Each of the measure of the radius of base of a cone and that of a sphere is 8 cm. Also, the volume of these two solids are equal. The slant height of the cone is
(1) 8 $\sqrt{17}$ cm (2) 4$\sqrt{17}$ cm
(3) 34 $\sqrt{2}$ cm (4) 34 cm.

4. A cone of height 15 cm and base diameter 30 cm is carved out of a wooden sphere of radius 15 cm. The percentage of wasted volume of the cone, at what height, above the base, is the section made?
(1) 6 cm (2) 8 cm
(3) 10 cm (4) 20 cm

9. The radius of the base and height of a right circular cone are in the ratio 5 : 12. If the volume of the cone is $314\frac{2}{7}$ cm^3, the slant height (in cm) of the cone will be
(1) 12 (2) 13
(3) 15 (4) 17

10. Two solid right cones of equal height and of radii r_1 and r_2 are melted and made to form a solid sphere of radius R. Then the height of the cone is
(1) $\frac{4R^2}{r_1^2+r_2^2}$ (2) $\frac{4R}{r_1+r_2}$
(3) $\frac{4R^3}{r_1^2+r_2^2}$ (4) $\frac{R^2}{r_1^2+r_2^2}$